Youssef Elhagrasy

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Research Interests

Algorithm-hardware co-design for efficient AI; compact data representations; accelerator memory systems and data movement; ASIC/RTL design; reliable and energy-efficient inference; Vision Transformer representation learning; server energy measurement and attribution.

Education

University of British Columbia (UBC), Vancouver, Canada
Bachelor of Applied Science (BASc), Engineering Physics
Specialization: Electrical and Computer Engineering — Sep 2022 – May 2027
Dean's Honour List; **GPA: 4.33/4.33**

Publications

- **Y. A. Elhagrasy**, Y. Abulnaga, D. H. An, and B. Quinton, "Combined Per-Kernel Weight Pruning and Activation Thresholds for Energy-Efficient DNN Inference," submitted to *MLSys*, 2027.
- **Y. A. Elhagrasy**, I. Hill, and A. Ivanov, "Adaptive Sequential Test Planning for Multi-Mechanism Reliability Qualification via Bayesian Monte Carlo Tree Search," *IEEE Transactions on Reliability*, accepted, 2026.
- R. Guerrero, **Y. A. Elhagrasy**, P. Moheban, I. Hill, and A. Ivanov, "Wear-out Sensor SoC Peripheral for Continuous Semiconductor Reliability Monitoring, Lifespan Prediction, and Failure Risk Management," *IEEE Transactions on Device and Materials Reliability*, under review, 2026.
- X. Xu et al., including **Y. A. Elhagrasy**, "Wafer-Scale Transfer of MXene Films with Enhanced Device Performance via 2D Liquid Intercalation," *Advanced Materials*, vol. 36, no. 51, 2024.

Industry Experience

Microsoft — Silicon Engineering Intern — May 2026 – Aug 2026
Cobalt RTL Design Team, PCIe/CXL Bridge Infrastructure

- Owned RTL features and led JasperGold formal signoff for a CHI-/AXI-compliant CXL retry/credit block, reaching **99% checker COI and 99% proof-core coverage**.
- Built agentic CDC/RDC and formal-environment workflows that reduced turnaround **5×** and cut bring-up from **2 days to 3 hours**.
- Improved three critical timing paths by up to **40 ps** with physical-design teams and root-caused a CSR parity architecture flaw that could have caused a reliability escape.

Microsoft — Silicon Engineering Intern — May 2025 – Aug 2025
Azure Hardware Systems & Infrastructure

- Characterized wafer-level test data across **10M+ datapoints** for custom silicon.
- Identified layout- and timing-related issues and provided feedback to physical design teams.
- Designed an internal **RAG + knowledge graph + LangGraph AI agent** for code retrieval and reasoning, achieving **3× speedup**.

Tesla — Silicon Systems Validation Intern — Sep 2025 – Jan 2026

- Built automated validation infrastructure in **C/C++ and Python** on Linux for approximately **50 functional, performance, and reliability scenarios**.
- Executed hundreds of regressions and performed firmware flashing, bootloader configuration, log analysis, and cross-layer HW/SW debugging on production automotive SoCs.

Research Experience

University of British Columbia — Undergraduate Researcher, Efficient DNN Inference — Jan 2026 – Present
Supervisor: Prof. Brad Quinton

- Developed a two-stage PyTorch framework combining learned per-kernel binary masks and activation thresholds to skip ineffectual multiply-accumulate operations.
- Reduced estimated VGG16-BN inference energy by **28.5%** with only **0.05% accuracy loss** on CIFAR-10; validated the energy model through SkyWater 130 nm ASIC synthesis and led the resulting first-author MLSys submission.

University of British Columbia — Undergraduate Researcher, Reliability & Test — Feb 2023 – Present
Supervisor: Prof. André Ivanov

- Formulated adaptive reliability qualification across BTI, electromigration, and TDDB as a sequential decision problem; built a Bayesian MCTS-SA/EKF planner that improved yield from **20.4% to 54.2% over 5,000 iterations**.
- Developed BayeShield, a risk-constrained Bayesian diagnosis framework spanning **249,389 reachable belief states** and **3 binding risk ceilings**; related first-author work was accepted by *IEEE Transactions on Reliability*.
- Designed an **IEEE 1500-compliant DFT wrapper** around a data-acquisition IP and verified serial/parallel scan access, functional isolation, and test-mode behavior using SystemVerilog and UVM.

University of British Columbia — Undergraduate Researcher, Server Energy Systems Sep 2026 – Present
Supervisor: Prof. Mohammad Shahrad

- Investigating server-level energy tracking and attribution for cloud and serverless workloads, connecting application activity with OS scheduling, resource allocation, and hardware energy measurements.
- Studying how fractional-CPU enforcement, concurrency, and workload duration affect resource utilization, performance, and energy accounting in multi-tenant systems.

UBC — Undergraduate Researcher, Vision Transformers Sep 2026 – Present
Supervisor: Prof. Abdelrahman Eldesokey

- Investigating representation learning in Vision Transformers, with emphasis on how patch-token features encode semantic and spatial structure and transfer under distribution shift.
- Exploring self-supervised ViT representations and evaluation methods for robust visual correspondence and fine-grained vision tasks.

KAUST — Transistor Undergraduate Researcher Sep 2023 – May 2024
Supervisors: Prof. Husam AlShareef, Prof. Xiangming Xu

- Fabricated and characterized transistors using **MoS_2 and MXenes** via 2D thin-film coating for transparent electronics.
- Designed experiments and analyzed electrical traces to study performance variability, achieving **¿1000% improvement** in key device metrics.
- Applied **Design of Experiments (DoE)**, ALD, and chemical/dry etching for systematic characterization.
- Co-authored the resulting paper published in *Advanced Materials*, vol. 36, no. 51, 2024.

TRIUMF — Firmware & Simulation Research Intern May 2023 – Aug 2023

- Developed and optimized **C++ simulation code** on Linux for particle trajectories, energy deposition, and detector response.
- Built Geant4-based simulation models for dark matter experiments in collaboration with TRIUMF, SNOLAB, and Fermilab.
- Documented and analyzed systematic error sources for the Scintillating Bubble Chamber (SBC).
- Selected as **1 of 3 students from 2000+ applicants**.

Selected Projects

Attention Accelerator: CUDA to RTL/FPGA 2026 – Present

- Built scaled dot-product attention across float64/Q8.8 Python models, optimized CUDA kernels, and two synthesizable SystemVerilog architectures.
- Replaced materialized attention with online softmax, reducing simulated RTL latency from **659 to 240 cycles** and intermediate storage from approximately **64 KB to 66 words** at $N = 128, D_V = 64$.
- Packaged the core as AXI4-Stream/DMA IP with performance counters and completion interrupts; optimized CUDA reached **3.86 TFLOP/s** and a **12.7× speedup** at $N = 4096, D = 128$.

Accelerator Memory-System Simulator 2026

- Built a CI-gated, approximately **2.6K-line Python simulator** for scratchpad banking, tensor layouts, memory traffic, and layout-persistent accelerator execution.
- Swept **6,720 configurations** and measured a **3.0× median traffic penalty** from mismatched tensor layouts, translating results into reproducible architecture recommendations.

Optimized Serial Adder 2024

- Designed a transistor-level datapath and FSM in Cadence Virtuoso using a 45 nm GPDK.
- Applied logical effort to achieve **2.4× throughput** (**58% lower critical-path delay**) and completed a DRC/LVS-clean layout.

Technical Skills

Languages: Python, C/C++, CUDA, SystemVerilog, Verilog, MATLAB, Tcl/Bash, LaTeX
ML/AI: PyTorch, TensorFlow, Vision Transformers, self-supervised learning, pruning/sparsity, MCTS, Bayesian estimation
Systems: Linux, accelerator simulation, memory/performance modeling, Nsight Compute, server energy measurement, Git/CI
Hardware: ASIC/VLSI, FPGA, AXI, fixed-point arithmetic, SVA, UVM, DFT, JasperGold, Cadence Virtuoso, Vivado, DRC/LVS

Honors & Awards

Faculty of Applied Science International Student of the Year (2×); Dean's Honour List (3×); Eric Vogt Scholar; IMES; OIS; Saudi RSI Top Student & Presenter